Energy-Guided Smoothness for Robust Graph Classification under Label Noise

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Abstract

Graph Neural Networks (GNNs) are vulnerable to label noise in graph classification, yet the mechanisms driving this vulnerability are poorly understood. We show robustness in this setting is governed by a property of the *learned representations*: the within-graph Dirichlet energy of node features. Energy decreases as the model captures genuine class structure, then rises sharply, mainly in high-frequency components, when the model begins to memorize corrupted graph labels. We give a theoretical justification for why this energy decrease is non-trivial in graph classification, combining a spectral identity at the dataset level with an entropy-based argument adapted from Qian et al. [2025] that shows uncontrolled energy growth disproportionately harms the very graphs most informative for classification. Building on this, we propose three convergent interventions, a spectral weight constraint (CE+W2), a Dirichlet-energy regularizer ($\mathcal{L}_{DE}$), and a robust loss (GCOD). Across eight benchmarks, including the full and modified ogbg-ppa, symmetric noise up to 60%, asymmetric noise, and a transfer to Cora node classification, all three methods improve robustness while preserving clean-data performance, with GCOD matching or exceeding OMG [Yin et al., 2023] at $\sim 70\times$ lower training overhead.

1 Introduction

Graph Neural Networks (GNNs) are now the workhorse of *graph classification*, the task of assigning a single label to an entire graph, with applications spanning molecular property prediction [Wale and Karypis, 2006], bioinformatics [Borgwardt et al., 2005], and social-network analysis [Yanardag and Vishwanathan, 2015a]. In real deployments, the labels attached to graphs are routinely noisy: assays disagree, annotators err, and weak supervision injects systematic mistakes. Although learning under label noise has been thoroughly studied for images [Algan and Ulusoy, 2021] and for node classification [Dai et al., 2021, Yuan et al., 2023a], graph classification under noisy labels is dramatically less explored [NT et al., 2019, Yin et al., 2023].

Graph classification is a fundamentally different setting. In node classification, label noise corrupts *individual node labels* on a single graph, and homophily, the tendency of connected nodes to share labels, is a structural prior that directly couples Dirichlet energy of the *label signal* to noise. In graph classification, label noise flips the label of an *entire graph*: nodes have no individual classification labels, and within-graph homophily/heterophily is largely orthogonal to the graph-level label. The standard tools and intuitions from node classification therefore do not transfer.

Our central observation is non-trivial. We measure the Dirichlet energy of *learned node representations*, not of labels. Empirically, this energy first decreases as the model captures genuine, transferable patterns. Then, when the model begins to memorize incorrect graph-level labels, the

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within-graph energy rises sharply, with the surge concentrated in the high-frequency spectrum. This is not a tautology: noise is injected only at the graph-label level, while node features and edges are unchanged. We show in Theorem 1 that, under a bi-Lipschitz assumption on the GNN, fitting noisy graph-level labels for two structurally similar graphs *forces* their within-graph Dirichlet energies to grow, via the Graph Poincaré inequality. The connection between graph-level label noise and within-graph spectral sharpening therefore emerges through training dynamics, not by definition.

Why does suppressing this energy growth help? Combining a dataset-level spectral identity (Proposition 1) with an entropy-based argument from Qian et al. [2025], we show in Section 6 that uncontrolled within-graph energy growth disproportionately damages *high-entropy graphs*, the ones carrying most discriminative signal at the pooling stage. A non-trivial decrease of within-graph Dirichlet energy thus simultaneously (i) prevents fitting graph-level label noise and (ii) preserves the high-entropy regions on which classification accuracy depends.

Contributions.

- A systematic study of *when* GNNs fit graph-level label noise (Section 4), isolating three failure modes: over-parameterization, low-order graphs, small training sets.
- Dirichlet energy as a diagnostic for graph-level noise memorization (Section 5), with a frequency decomposition localizing the surge to high-frequency components.
- *Theoretical grounding* for why a non-trivial energy decrease prevents noise fitting in graph classification (Section 6): a spectral identity (Proposition 1), a Poincaré–bi-Lipschitz lower bound (Theorem 1), and an entropy-based asymmetry result (Corollary 1). The connection is non-trivial for graph classification, unlike its trivial node-classification analogue.
- Three convergent interventions (Section 7): a spectral weight constraint (CE+W2), explicit Dirichlet regularization ($\mathcal{L}_{DE}$), and a centroid-based loss GCOD. Mechanistically distinct, they all reduce energy and improve robustness.
- Eight benchmarks under symmetric, asymmetric and high-rate (60%) noise, full and modified ogbg-ppa, and a Cora transfer. GCOD matches OMG [Yin et al., 2023] and SOP [Liu et al., 2022] at $\sim$70$\times$ lower compute.

2 Related Work

Learning under label noise. Robust loss functions [Ghosh et al., 2017, Patrini et al., 2017], sample re-weighting [Liu and Tao, 2015, Liu et al., 2022], early-learning regularizers [Arpit et al., 2017], and small-loss filtering [Gui et al., 2021] dominate the image-classification literature. Wani et al. [2023] introduce NCOD, a centroid-based outlier-discounting loss that we adapt to graphs.

Noisy-label learning on graphs. Most prior work concerns *node* classification, exploiting homophily, contrastive losses, or pseudo-labels [Dai et al., 2021, Yuan et al., 2023a,b, Li et al., 2024, Kang et al., 2018]. For *graph* classification, NT et al. [2019] propose a surrogate loss that discards estimated-noisy labels under specific assumptions, and OMG [Yin et al., 2023] combines contrastive learning, MixUp [Lim et al., 2022], and curriculum filtering. We differ in two ways: (i) we provide a spectral/energy-based explanation of *why* GNNs fit graph-level noise, rather than only mitigating it, and (ii) our practical method (GCOD) achieves better or comparable robustness at a fraction of the compute.

Dirichlet energy and over-smoothing. Dirichlet energy quantifies signal smoothness on graphs [Zhou and Schölkopf, 2005]. Existing analyses establish that GNNs act as low-pass filters whose energy decays exponentially with depth under spectral conditions [NT and Maehara, 2019, Cai and Wang, 2020, Oono and Suzuki, 2020, Rusch et al., 2023], and that weight-matrix eigenvalues control attraction/repulsion of node features [Giovanni et al., 2023]. Energy-based regularizers [Zhou et al., 2021, Bo et al., 2021, Chen et al., 2023] address *too little* energy (over-smoothing) on *node* classification. We address the opposite regime: *too much* within-graph energy caused by graph-level label noise. Qian et al. [2025] recently show that, for graph classification specifically, over-smoothing in *high-entropy* regions is the most damaging; we leverage this insight (Corollary 1) to argue why suppressing energy growth is essential under graph-level noise. A more comprehensive related-work discussion is in Appendix A.

3 Background

Notation. A graph is $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ with $N = |\mathcal{V}|$, degree matrix $\mathbf{D}$, adjacency $\mathbf{A} \in \{0,1\}^{N\times N}$, feature matrix $\mathbf{X} \in \mathbb{R}^{N\times m}$, combinatorial Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ (with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_N$) and normalized Laplacian $\mathbf{\Delta} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2}$. We write $\lambda_2(\mathcal{G})$ for $\lambda_2(\mathbf{L})$ throughout and use $\mathbf{\Delta}$ only when explicitly stated. In *graph classification*, the dataset is $\mathcal{D} = \{(\mathcal{G}^i, \mathbf{y}_i)\}_{i=1}^n$, $\mathbf{y}_i \in \{0,1\}^{|C|}$ the one-hot graph label; we write c_i for the class. Under message passing, $\mathbf{H}_i^{l+1} = \mathrm{UP}_{\mathbf{\Omega}_l}(\mathbf{H}_i^l, \mathrm{AGGR}_{\mathbf{W}_l}(\mathbf{H}_i^l, \mathbf{A}^i))$, with $\mathbf{H}_i^0 \equiv \mathbf{X}^i$ and $\mathbf{Z}^i \equiv \mathbf{H}_i^L$. A permutation-invariant readout $f_\theta : \mathbb{R}^{N_i\times m'} \to \mathbb{R}^{|C|}$ maps node features to class probabilities $\hat{\mathbf{y}}_i$. Under noisy labels, the observed c_i may differ from the ground truth; the test set is clean.

Dirichlet energy on graphs. For graph $\mathcal{G}^i$ with representation $\mathbf{Z}^i \in \mathbb{R}^{N_i\times m'}$ we use the combinatorial Dirichlet energy

$$E^{\mathrm{dir}}(\mathbf{Z}^i) = \sum_{(u,v)\in\mathcal{E}^i} \left\|\mathbf{Z}_u^i - \mathbf{Z}_v^i\right\|_2^2 = \mathrm{tr}\big((\mathbf{Z}^i)^\top \mathbf{L}^i \mathbf{Z}^i\big). \tag{1}$$

Small E^{dir} means smooth (low-frequency) representations within the graph; large E^{dir} means sharpening (high-frequency). The symmetrically-normalized variant $E^{\mathrm{dir}}_{\mathrm{sym}}(\mathbf{Z}) = \sum_{(u,v)} \|\mathbf{Z}_u/\sqrt{d_u} - \mathbf{Z}_v/\sqrt{d_v}\|_2^2 = \mathrm{tr}(\mathbf{Z}^\top \mathbf{\Delta}\mathbf{Z})$, used in some over-smoothing analyses [NT and Maehara, 2019, Oono and Suzuki, 2020], is equivalent up to degree-bounded constants and our qualitative claims hold for either; we use (1) because it pairs cleanly with the Poincaré inequality of Section 6. To track behavior across a class-restricted subset $\mathcal{S}$ of training graphs we use

$$E^{\mathrm{dir}}_{\mathrm{set}}(\mathcal{S}) = \frac{1}{|\mathcal{S}|} \sum_{\mathcal{G}^i\in\mathcal{S}} E^{\mathrm{dir}}(\mathbf{Z}^i). \tag{2}$$

Crucially, (1) is computed on *learned representations*, not on labels. Within-graph homophily of features therefore evolves with training, decoupled from the graph-level label of $\mathcal{G}^i$.

4 When GNNs Fit Graph-Label Noise

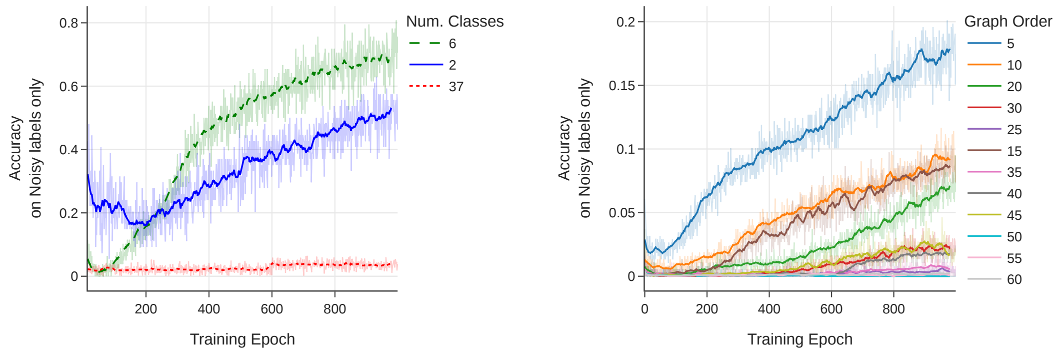
GNNs trained with standard cross-entropy (CE) on graph classification tasks are not uniformly vulnerable to label noise. On the full ogbg-ppa [Szklarczyk et al., 2018a] (37 classes, the standard OGB benchmark), even 40% symmetric noise barely degrades CE accuracy, because the model is under-parameterized for the full task and behaves robustly [Zhang et al., 2021]. To expose the noise-fitting regime, we control three axes: **(F1)** number of classes (proxy for over-parameterization), **(F2)** graph order (size of each graph), and **(F3)** training-set size. We use a 5-layer GIN [Xu et al., 2019] with 300 hidden units throughout, unless stated.

In all three regimes (Fig. 1 and Appendix D), GNNs progressively memorize noisy graph-level labels. This narrows the question to: what changes in the network's internal representations during memorization, and can this change be controlled?

5 Dirichlet Energy as a Diagnostic for Noise Fitting

We track $E^{\mathrm{dir}}_{\mathrm{set}}(\mathcal{D}_c)$, the within-graph energy averaged over training graphs of class c. Across architectures and datasets (Fig. 2), the diagnostic is consistent: $E^{\mathrm{dir}}_{\mathrm{set}}$ falls during the early "learning" phase, then rises precisely when the model begins to fit noisy graphs (CE-20% in Fig. 2b), with 40% noise yielding a strictly larger end-of-training energy than 10% noise (Appendix F).

Energy is sensitive to noise specifically through high-frequency components. A natural concern is that energy also rises during clean-data overfitting [Yan et al., 2022]; we confirm this in Appendix F but show the noise-driven surge is markedly steeper. To localize the surge spectrally, we use the HLFF-GNN decomposition [Xu et al., 2024] to split representations into low-frequency (Z_1) and high-frequency (Z_2) components. On ENZYMES with 30% noise, $E^{\mathrm{dir}}(Z_1)$ remains essentially unchanged, while $E^{\mathrm{dir}}(Z_2)$ rises monotonically, and slope/variance statistics of $E^{\mathrm{dir}}(Z_2)$ provide an early warning for label noise (Appendix E). HLFF-GNN is used purely as an analytical tool; the same total-energy surge appears in standard GIN/GCN.



(a) Effect of class count.

(b) Effect of graph order.

Figure 1: Training accuracy on *noisy labels only*: a higher curve means more memorization. (a) Reducing the number of PPA classes (with 20% symmetric noise) makes a fixed GIN over-parameterized, accelerating noise fitting. (b) Smaller average graph order (synthetic data, 35% noise) leaves less internal structure for averaging, enabling memorization.

Why energy is more useful than a generic overfitting signal. Standard noise-detection signals require a *clean* held-out set: validation loss is uninformative when validation labels follow the same noise distribution as training (the noisy validation loss continues to decrease as the model fits noise, as we observe in our setting); train–test gap is unobservable at training time; small-loss heuristics need a calibration regime. Within-graph Dirichlet energy is computed entirely on the training graphs and their representations, so it remains a valid noise-fitting indicator under fully-noisy supervision. Beyond replacing the missing clean signal, it tells you *how* the network is failing: which spectral components drive the surge. This mechanistic specificity is what turns a diagnostic into a design principle, and motivates the three interventions in Section 7, each of which targets the same harmful spectral behavior in a distinct way.

6 Why a Non-Trivial Energy Decrease Prevents Noise Fitting

We show that under standard regularity, within-graph Dirichlet energy must *grow* when a GNN fits graph-level label noise, and the growth is absorbed disproportionately by structurally rich graphs. The argument combines three pieces: a dataset-level spectral identity (Proposition 1), the Graph Poincaré inequality (Lemma 1), and a bi-Lipschitz lower bound on the GNN (Lemma 2) [Chuang and Jegelka, 2022, Davidson and Dym, 2025, Juvina et al., 2024], yielding Theorem 1.

6.1 Three Building Blocks

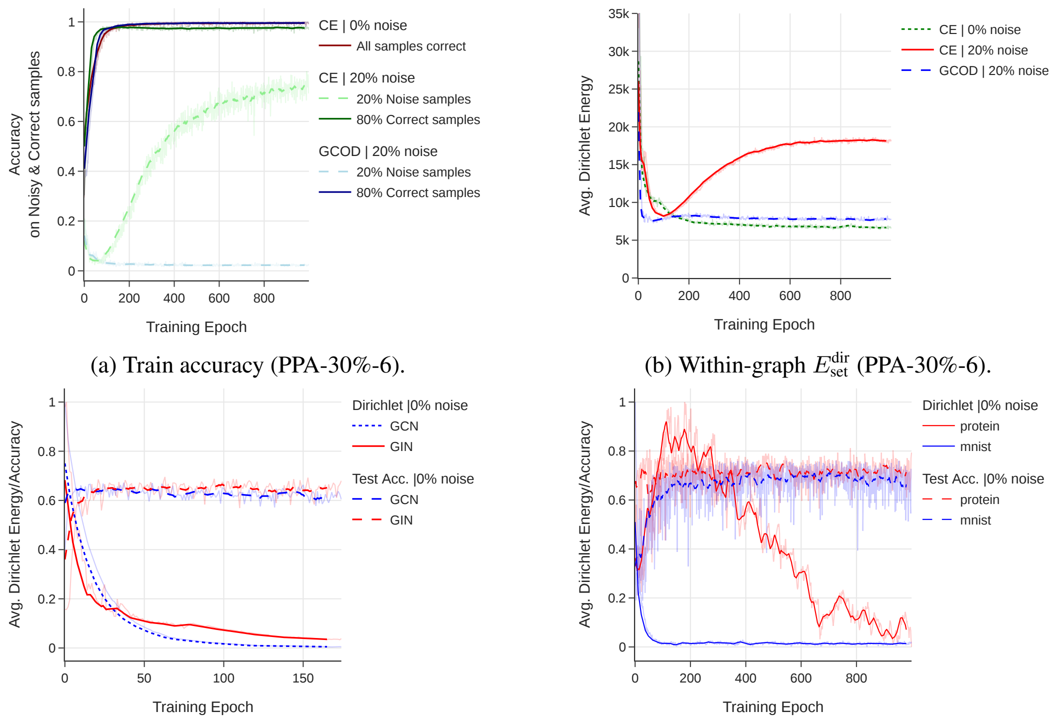
Proposition 1 (Dataset-level Dirichlet energy)**.** *Let $\mathcal{D} = \{\mathcal{G}^1, \dots, \mathcal{G}^n\}$ with representations $\mathbf{Z}^i \in \mathbb{R}^{N^i \times m'}$ and adjacencies $\mathbf{A}^i$. Define the block-diagonal graph $\bar{\mathcal{G}} = (\bar{\mathbf{Z}}, \bar{\mathbf{A}})$ with $\bar{\mathbf{Z}} = [\mathbf{Z}^1 \| \dots \| \mathbf{Z}^n] \in \mathbb{R}^{N_{tot} \times m'}$ (vertical concatenation) and $\bar{\mathbf{A}} = \mathrm{blkdiag}(\mathbf{A}^1, \dots, \mathbf{A}^n)$, with associated combinatorial Laplacian $\bar{\mathbf{L}} = \mathrm{blkdiag}(\mathbf{L}^1, \dots, \mathbf{L}^n)$. Then*

$$E^{dir}_{set}(\mathcal{D}) = \tfrac{1}{|\mathcal{D}|} E^{dir}(\bar{\mathbf{Z}}) = \tfrac{1}{|\mathcal{D}|} \sum_{r=1}^{m'} \sum_{u=1}^{N_{tot}} \bar{\lambda}_u (\bar{\psi}_u^\top \bar{\mathbf{Z}}_{:,r})^2, \tag{3}$$

where $\{(\bar{\lambda}_u, \bar{\psi}_u)\}$ are eigenpairs of $\bar{\mathbf{L}}$, and the spectrum decomposes as $\bar{\Lambda} = \bigcup_i \Lambda^i$ with each eigenvector supported on a single connected component $\mathcal{G}^i$.

The proof (Appendix B) follows from the block-diagonal structure of $\bar{\mathbf{L}}$. The key structural fact: *each high-frequency eigenvector of $\bar{\mathbf{L}}$ is owned by exactly one graph*, so sharpening along it raises only that graph's energy.

Lemma 1 (Graph Poincaré, vector-valued form)**.** *For any feature matrix $\mathbf{Z} \in \mathbb{R}^{N \times m'}$ on a connected graph $\mathcal{G}$ with combinatorial Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ and second-smallest eigenvalue $\lambda_2(\mathcal{G}) > 0$, let*



(a) Train accuracy (PPA-30%-6).

(b) Within-graph $E^{\text{dir}}_{\text{set}}$ (PPA-30%-6).

(c) Energy/accuracy across architectures.

(d) Across datasets.

Figure 2: Within-graph Dirichlet energy of learned representations *rises during noise memorization*, across architectures (GIN/GCN) and datasets (PPA, MUTAG, MNIST, PROTEINS). Under clean labels, $E^{\text{dir}}_{\text{set}}$ steadily decays as accuracy improves; under noise, it follows the same decay until the model starts fitting noisy graphs, after which it sharply rises. GCOD suppresses this surge.

$\bar{\mathbf{z}} = \frac{1}{N}\sum_v \mathbf{Z}_v$. *Then*

$$\sum_{v=1}^{N} \|\mathbf{Z}_v - \bar{\mathbf{z}}\|_2^2 \;\le\; \frac{1}{\lambda_2(\mathcal{G})}\, E^{dir}(\mathbf{Z}). \tag{4}$$

(4) bounds within-graph feature variance by Dirichlet energy scaled by the spectral gap; proof by Courant–Fischer per feature dimension (Appendix B).

Lemma 2 (Bi-Lipschitz lower bound for a GIN block with 2-layer combine, adapted from Chuang and Jegelka, 2022, Davidson and Dym, 2025, Juvina et al., 2024)**.** *Let $h_\theta(\mathbf{X}) = \sigma_2\big(\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\,\mathbf{W}_2\big)$ be a single GIN aggregation $\mathbf{M} = (1+\varepsilon)\mathbf{I} + \mathbf{A} \in \mathbb{R}^{N\times N}$ followed by a two-layer MLP with weights $\mathbf{W}_1 \in \mathbb{R}^{m\times m'}$ and $\mathbf{W}_2 \in \mathbb{R}^{m'\times m''}$ acting on the feature dimension, with the widening regime $m \le m' \le m''$ (so the linear maps have full column rank). Assume $\sigma_{\min}(\mathbf{M}) \ge \kappa > 0$ (e.g., $1+\varepsilon > |\lambda_{\min}(\mathbf{A})|$), $\sigma_{\min}(\mathbf{W}_k) \ge m_k > 0$, and that σ_1, σ_2 are leaky-ReLU with slope $\alpha > 0$ (or ReLU operated with a strict margin $\delta > 0$ along the line segment $\gamma(t) = \mathbf{X} + t\Delta\mathbf{X}$). Then*

$$\|h_\theta(\mathbf{X}+\Delta\mathbf{X}) - h_\theta(\mathbf{X})\|_F \;\ge\; c_{lower}\,\|\Delta\mathbf{X}\|_F, \qquad c_{lower} = \alpha^2 m_1 m_2 \kappa. \tag{5}$$

A second aggregation step (turning the block into a true two-layer GIN $\sigma_2(\mathbf{M}\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\mathbf{W}_2)$) replaces c_{lower} by $\alpha^2 m_1 m_2 \kappa^2$; the qualitative content of the lemma is unchanged.

Proof: singular-value chain rule on the vectorized Jacobian $\mathbf{J} = \mathbf{D}_2(\mathbf{W}_2^\top \otimes \mathbf{I}_N)\mathbf{D}_1(\mathbf{W}_1^\top \otimes \mathbf{M})$ (Appendix B). The lemma encodes that the GIN does not collapse meaningful feature differences, a property the model exploits when fitting noisy labels.

6.2 Main Theorem: Noise Fitting Forces Energy Growth

Consider two graphs $\mathcal{G}^i, \mathcal{G}^j$ with $N^i = N^j = N$ and similar pooled descriptors $\mathbf{g}_i \approx \mathbf{g}_j$ on the clean distribution (the same-N assumption is for notational simplicity; unequal N in Appendix B).

Suppose label noise flips c_j to $c_i' \neq c_i$, forcing the network to output different graph-level labels. Let $\mathbf{h}_i \in \mathbb{R}^{N\times m'}$ be the post-message-passing signal of $\mathcal{G}^i$, $\mathbf{g}_i = \frac{1}{N}\sum_v \mathbf{h}_{i,v}$, and $\bar{\mathbf{h}}_i = \mathbf{1}_N \mathbf{g}_i^\top$.

Theorem 1 (Graph-level noise fitting forces E^{dir} to grow)**.** *Under the hypotheses of Lemmas 1–2 and $N^i = N^j = N$,*

$$\|\mathbf{h}_i - \mathbf{h}_j\|_F \;\le\; \sqrt{N}\,\|\mathbf{g}_i - \mathbf{g}_j\|_2 + \sqrt{\frac{E^{dir}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\frac{E^{dir}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}}. \tag{6}$$

In particular, if the noisy training objective combined with an L-Lipschitz readout forces a graph-level output gap of at least Δ_{out}, so that $\|\mathbf{h}_i - \mathbf{h}_j\|_F \ge \Theta_{noise} := L^{-1}\Delta_{out}$, while the structural similarity gives $\|\mathbf{g}_i - \mathbf{g}_j\|_2 \le \varepsilon$, then

$$\sqrt{\frac{E^{dir}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\frac{E^{dir}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}} \;\ge\; \Theta_{noise} - \sqrt{N}\,\varepsilon. \tag{7}$$

Why this is non-trivial. (7) *lower-bounds* the within-graph Dirichlet energies. For structurally similar graphs that must be mapped to different labels, $\mathbf{g}_i \approx \mathbf{g}_j$, so the gap Θ_{noise} cannot come from the means; by Poincaré, the only remaining route is the within-graph variance E^{dir}/λ_2. Energy growth is a *mathematical consequence* of noise fitting, not a coincidence. Lemma 2 ensures the gap cannot be obtained for free by collapsing inputs ($c_{\text{lower}} > 0$). The unequal-N case (zero-padded permutation matching) is treated in Appendix B. This bound has no node-classification analog, where labels live on the same nodes whose smoothness E^{dir} measures.

6.3 Asymmetric Burden on High-Entropy Graphs

(7) only constrains the *sum* of $\sqrt{E^{\text{dir}}(\mathbf{Z}^i)/\lambda_2(\mathcal{G}^i)}$ across the two graphs. We next show that the structural-entropy ranking of Qian et al. [2025] dictates which graph absorbs the burden.

For a graph $\mathcal{G}$, the structural Shannon entropy is $H(\mathcal{G}) = -\sum_i \frac{d_i}{\sum_j d_j}\log\frac{d_i}{\sum_j d_j}$. Theorems 1–2 of Qian et al. [2025] prove that more nodes (at fixed sparsity) and denser connectivity (at fixed $|V|$) both strictly increase H.

Lemma 3 (Spectral counting)**.** *The combinatorial Laplacian $\mathbf{L}^i$ has at most N^i distinct eigenvalues, contained in $[0,\, \max_{(u,v)\in\mathcal{E}^i}(d_u + d_v)]$, hence in $[0,\, 2d_{\max}(\mathcal{G}^i)]$ [Chung, 1997]. Equivalently, the normalized spectrum $\Lambda(\mathbf{\Delta}^i) \subset [0,2]$ with $\lambda_{\max}(\mathbf{\Delta}^i) = 2$ iff $\mathcal{G}^i$ has a bipartite component. Therefore, when sharpening lifts spectral mass onto eigenvectors $\bar{\psi}_u$ of $\bar{\mathbf{L}}$ with $\bar{\lambda}_u \ge \tau$ for some threshold τ, the share allocated to $\mathcal{G}^i$ is bounded above by the count $n^i(\tau) = |\{j : \lambda_j^i \ge \tau\}|$, which is non-decreasing in N^i and in connection density (by Cauchy interlacing under vertex/edge addition).*

Corollary 1 (Asymmetric burden)**.** *If $H(\mathcal{G}^j) \ge H(\mathcal{G}^i)$ via either of the conditions in Qian et al. [2025, Thm. 1–2], then the count of high-frequency eigenvalues of $\mathcal{G}^j$ above any threshold τ is at least that of $\mathcal{G}^i$, so the energy increase forced by Theorem 1 is preferentially absorbed by $\mathcal{G}^j$. This collapses the node-distribution distinctness of $\mathcal{G}^j$ [Qian et al., 2025, Sec. 3.3], the very property pooled graph classifiers rely on.*

Implication. Theorem 1 says noise fitting must raise energy somewhere; Corollary 1 says it raises it preferentially in the graphs the classifier most needs to distinguish. Suppressing within-graph energy growth therefore blocks the noise-fitting path while preserving the classification signal, an alignment that has no node-classification analog. This also explains why the surge of $E^{\text{dir}}(Z_2)$ under noise (Section 5) is steeper than under clean overfitting: noise fitting on similar graphs is structurally constrained to high-frequency, high-entropy directions, while clean overfitting can spread across the full spectrum.

7 Three Convergent Methods

The theory of Section 6 identifies three distinct levers to suppress harmful within-graph energy growth: the *spectrum of the weight matrices* (architectural), the *energy of representations* (regularization), and the *loss function* (sample re-weighting). We design one intervention for each, and use them as convergent evidence for the energy framework.

7.1 Method 1: Spectral Constraint on Weight Matrices (CE+W2)

Giovanni et al. [2023] show that for linear graph convolutions with *symmetric* $\mathbf{W} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top$, the gradient flow of E^{dir} decomposes along eigenvectors of $\mathbf{W}$: positive λ_k smooth, negative λ_k sharpen. With Proposition 1, this argues for projecting $\mathbf{W}_l^2$ onto its positive eigenspace. For non-symmetric $\mathbf{W}_l^2$ the dichotomy no longer applies cleanly (complex eigenvalues, rotational flow), so we split $\mathbf{W}_l^2 = \mathbf{S}_l + \mathbf{N}_l$ into symmetric ($\mathbf{S}_l$) and skew-symmetric ($\mathbf{N}_l$, purely imaginary spectrum) parts and project only $\mathbf{S}_l$.

Procedure. After each gradient step, eigendecompose $\mathbf{S}_l = \boldsymbol{\Phi}_l \boldsymbol{\mu}_l \boldsymbol{\Phi}_l^\top$, set $\boldsymbol{\mu}_l \to [\boldsymbol{\mu}_l]^+$, and reconstruct $\mathbf{W}_l^{2+} = \boldsymbol{\Phi}_l [\boldsymbol{\mu}_l]^+ \boldsymbol{\Phi}_l^\top + \mathbf{N}_l$ (detached from the gradient graph; Appendix C.4). **CE+W2** is CE training with this projection; no clean validation data or extra hyperparameters. The per-epoch eigendecomposition is costly and can destabilize convergence (Appendix C.4), motivating the alternatives below.

7.2 Method 2: Explicit Dirichlet-Energy Regularization ($\mathcal{L}_{DE}$)

A more direct lever is to penalize $E^{\text{dir}}(\mathbf{Z}^i)$ above a threshold U_{c_i} that is class-aware:

$$\mathcal{L}_{DE}(\mathcal{D}) = \frac{1}{N} \sum_{i=1}^{N} \left[\max(0, E^{\text{dir}}(\mathbf{Z}^i) - U_{c_i})\right]^2, \quad \mathcal{L} = \mathcal{L}_{CE} + \lambda \, \mathcal{L}_{DE}. \tag{8}$$

We instantiate the threshold in two ways. **Class-specific:** U_c is recomputed every epoch as the mean E^{dir} over a small clean validation set restricted to class c. This adapts to class-dependent baseline energies (e.g. a small molecular graph's natural energy is different from a social graph's), but requires a clean held-out set. **Fixed:** $U_c = U$ globally, a single hyperparameter that does not require any clean validation data, at the cost of mild over-smoothing on clean labels. Both variants improve robustness; the class-specific variant additionally improves *clean-label* accuracy by acting as a generalization-aware regularizer.

7.3 Method 3: Graph Centroid Outlier Discounting (GCOD)

GCOD is our practical recommendation. It extends NCOD [Wani et al., 2023] (originally an image-classification loss) to graph classification, with two design changes that are necessary in the graph-classification regime: a third KL term that disentangles clean from noisy graphs via per-sample trainable parameters $\mathbf{u} \in [0,1]^n$, and feedback through the current training accuracy a_{train} that prevents premature outlier discounting. For batch B with predictions $f_\theta(\mathbf{Z}_B)$, hard predictions $\hat{\mathbf{y}}_B$, soft labels $\tilde{\mathbf{y}}_B$ (centroid-based, see Appendix C.6), and observed labels $\mathbf{y}_B$:

$$\mathcal{L}_1 = \mathcal{L}_{CE}\big(f_\theta(\mathbf{Z}_B) + a_{\text{train}} \, \text{diag}_{\text{vec}}(\mathbf{u}_B) \, \mathbf{y}_B, \ \tilde{\mathbf{y}}_B\big), \tag{9}$$

$$\mathcal{L}_2 = \frac{1}{|C|} \big\| \hat{\mathbf{y}}_B + \text{diag}_{\text{vec}}(\mathbf{u}_B) \, \mathbf{y}_B - \mathbf{y}_B \big\|^2, \tag{10}$$

$$\mathcal{L}_3 = (1 - a_{\text{train}}) \, D_{KL}\big\{\mathfrak{L}, \ \sigma(-\log \mathbf{u}_B)\big\}, \tag{11}$$

with $\mathfrak{L} = \log \sigma(\text{diag}_{\text{mat}}(f_\theta(\mathbf{Z}_B)\mathbf{y}_B^\top))$. Updates are split: $\theta \leftarrow \theta - \alpha \nabla_\theta(\mathcal{L}_1 + \mathcal{L}_3)$, $\mathbf{u} \leftarrow \mathbf{u} - \beta \nabla_{\mathbf{u}} \mathcal{L}_2$. Empirically, GCOD reduces $E^{\text{dir}}_{\text{set}}$ during training (Fig. 2) without explicitly minimizing it: by down-weighting graphs flagged as noisy via large u_i, it removes the only gradient pressure that would push the network toward high-frequency directions in the first place. GCOD *is not* a theoretically derived energy minimizer. Instead, the theory predicts that any sample-level mechanism that suppresses noise-driven gradients will reduce $E^{\text{dir}}_{\text{set}}$, which is exactly the convergent evidence we observe.

8 Experiments

Setup. We evaluate on ogbg-ppa [Szklarczyk et al., 2018a] (full and a controlled 30%/6-class subset), MUTAG, PROTEINS, ENZYMES, IMDB-BINARY, REDDIT-MULTI-12K, MSRC-21, MNIST-graph, and Cora (node classification, transfer test). Backbones are GIN [Xu et al., 2019] and GCN [Kipf and Welling, 2017]; full hyperparameters are in Appendix C. Baselines: standard CE; SOP [Liu et al., 2022], a strong noise-robust loss from image classification; OMG [Yin et al., 2023], the leading graph-classification noise method. All numbers are mean±std over 4 runs.

Table 1: Test accuracy on the PPA-30%-6 subset (5-layer GIN, mean±std over 4 runs). **Best** is highlighted in red and bold, second-best in blue. "CE clean only" shows accuracy on the clean subset of the noisy data, as a reference upper bound. Class-specific $\mathcal{L}^*_{DE}$ requires a clean validation set.

Noise	Method	Test Acc.		Train Acc.	
		Best	Last	Best	Last
0%	CE	96.25 ± 0.05	91.25	100.0	99.33
	GCOD	96.65 ± 0.52	93.25	99.23	99.03
	CE+W2	96.50 ± 1.12	86.50	99.76	99.29
	$\mathcal{L}_{DE}$ (Fixed)	96.58 ± 0.27	85.70	99.26	98.29
	$\mathcal{L}^*_{DE}$ (Class-spec.)	96.96 ± 0.04	88.67	99.41	98.67
20%	CE (clean only)	96.15 ± 0.09	90.66	84.50	84.07
	CE	88.66 ± 0.16	62.58	94.98	93.45
	SOP	91.01 ± 0.44	85.50	79.17	77.64
	GCOD	93.91 ± 0.26	92.58	80.11	79.52
	CE+W2	89.83 ± 1.23	76.66	84.90	83.68
	$\mathcal{L}_{DE}$ (Fixed)	89.69 ± 0.57	80.25	78.07	70.14
	$\mathcal{L}^*_{DE}$	92.34 ± 0.41	80.92	84.55	83.73
40%	CE (clean only)	95.08 ± 0.13	80.44	68.15	67.05
	CE	82.08 ± 0.22	51.83	82.47	78.88
	SOP	82.33 ± 1.32	65.41	58.90	57.68
	GCOD	93.88 ± 0.04	91.08	65.09	64.31
	CE+W2	88.25 ± 1.13	56.33	68.78	65.88
	$\mathcal{L}_{DE}$ (Fixed)	88.58 ± 0.75	77.12	61.08	54.53
	$\mathcal{L}^*_{DE}$	88.55 ± 0.11	71.83	68.98	67.80

Why a controlled PPA subset? On the full 37-class PPA, all methods are under-parameterized for the task, and CE itself is robust (Section 4); a noise-robustness method cannot help where the baseline does not fail. We use a 30%/6-class PPA subset as a *controlled* testbed where over-parameterization triggers noise fitting, and we additionally report the full-PPA result to show our methods do no harm there. Generality claims are grounded in the seven additional benchmarks.

8.1 Main Result: Controlled PPA Subset

Table 1 reports the full ablation on the controlled PPA subset. GCOD is best at every noise level and, crucially, has the smallest gap between best and final test accuracy, indicating that it does not overfit late in training. CE+W2 and $\mathcal{L}_{DE}$ both improve robustness over CE, confirming the convergent-evidence prediction of Section 6. Under clean labels, $\mathcal{L}^*_{DE}$ even outperforms CE, a direct effect of the class-aware energy threshold.

8.2 Generalization Across Datasets and Noise Types

Table 2 reports GCOD across seven benchmarks beyond PPA. GCOD consistently improves over CE under both 20% symmetric and 40% asymmetric noise, and matches CE on REDDIT where the baseline is already saturated by class imbalance. Section 8 of Theorem 1's prediction (energy growth scales with the demanded Θ_{noise}) is consistent with the empirical observation that gains grow with the noise rate (Appendix F).

Full ogbg-ppa. Under the full 37-class benchmark, all methods including CE behave robustly to noise (Section 4); we therefore use it only as a do-no-harm check. Detailed results are in Appendix H.

Table 2: GCOD matches or beats CE across diverse datasets. Mean over 4 runs; ±std reported where available. *Sym*$_{20}$*:* 20% symmetric noise. *Asym*$_{40}$*:* 40% asymmetric noise. The Asym$_{40}$ GCOD column reports the std deviations from [Wani et al., 2023]'s rebuttal protocol.

Dataset	**0% CE**	**Sym$_{20}$ CE**	**Sym$_{20}$ GCOD**	**Asym$_{40}$ CE**	**Asym$_{40}$ GCOD**
PROTEINS	81.16	76.23	79.38	72.90	76.19 ± 1.22
MNIST	72.69	66.30	71.26	71.15	72.61 ± 0.43
ENZYMES	73.33	64.16	68.69	65.80	69.81 ± 0.98
IMDB-B	76.50	75.00	75.40	68.18	72.89 ± 0.91
MUTAG	94.73	89.47	90.01	91.16	93.19 ± 0.38
REDDIT	48.15	45.05	44.98	48.01	47.94 ± 2.87
MSRC-21	96.69	90.26	94.69	94.39	95.57 ± 0.32

Table 3: OMG protocol (Yin et al., 2023). Percentage gain over CE.

Dataset	**OMG**	**GCOD**
MUTAG	0.061	0.062
IMDB-B	0.047	0.049
PROTEINS	0.039	0.041

Table 4: Relative training time (CE-GIN = 1.00).

Method	**Runtime**
CE-GIN	1.00
SOP	1.048
GCOD	**1.029**
CE+W2	1.33
OMG [Yin et al., 2023]	≈ 3.00

Comparison with OMG. Under the OMG [Yin et al., 2023] protocol (Table 3), GCOD matches OMG's accuracy (within 0.1–0.2 pp) while requiring *∼70× less* training overhead (Table 4): OMG adds ≈200% runtime relative to CE-GIN, while GCOD adds only ≈2.9%. The accuracy-per-FLOP advantage is substantial and matters for any deployment beyond toy benchmarks.

Cross-setting transfer to node classification (Cora). As a stress test of generality, we apply GCOD and CE+W2 unchanged to node classification on Cora under 20% symmetric noise, against 13 node-specific baselines (NRGNN, RTGNN, GNN-Cleaner, ERASE, CR-GNN, and others; full table in Appendix I). On both GAT and GCN backbones, our methods rank in the top half despite not being designed for node classification: CE+W2 reaches 0.8157 (rank #2) on GAT and 0.7247 (rank #5) on GCN; GCOD reaches 0.7910 (rank #4) on GAT and 0.7403 (rank #4) on GCN. The fact that an energy-based regularizer that does not access neighbor labels or homophily can outperform specialized node-classification methods is direct evidence that the perspective captures a general property of GNN training under label noise rather than a graph-classification-specific quirk.

8.3 Convergent Evidence for the Energy Story

The three methods are mechanistically distinct (CE+W2 architectural, $\mathcal{L}_{DE}$ regularization, GCOD sample re-weighting), yet all (i) reduce $E^{\text{dir}}_{\text{set}}(\mathcal{D}_c)$ during training (Fig. 2, Appendix F) and (ii) improve robustness, exactly the convergence Theorem 1 and Corollary 1 predict.

9 Conclusion

GNN robustness to graph-level label noise is governed by within-graph Dirichlet energy of learned representations. The connection is non-trivial in graph classification, requires an entropy-based argument that has no node-classification analog, and yields three convergent interventions (CE+W2, $\mathcal{L}_{DE}$, GCOD) that all improve robustness across eight benchmarks while preserving clean-label performance. GCOD requires no clean validation data, adds only ≈3% training overhead, and matches OMG at ∼70× lower compute. **Limitations:** no closed-form noise–energy characterization yet; reliance on synthetic noise; no standardized real-world noisy graph-classification benchmarks. **Broader impact:** robust graph learning is most useful in chemistry and biology, where annotation errors are systematic; methods like GCOD that work without clean validation reduce deployment cost.

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A Additional Related Work

A.1 Learning under Label Noise

Beyond robust losses, additional families include sample relabelling [Arazo et al., 2019, Reed et al., 2015], two-network co-teaching and divide-and-mix approaches [Han et al., 2018, Li et al., 2020, Kim et al., 2023], and regularization through MixUp [Zhang et al., 2017]. Reweighting methods adapt sample importance via training dynamics or held-out probes [Liu and Tao, 2015, Pleiss et al., 2020]. Most of these were developed for image classification; their direct transfer to graph classification is non-obvious because graphs lack the strong feature regularities of images.

A.2 Graph Learning under Noise

Most graph-related noise work targets node classification, exploiting either homophily (RS-GNN [Dai et al., 2022], NRGNN [Dai et al., 2021], RTGNN [Yuan et al., 2023a], pairwise interactions [Du et al., 2023]) or self-supervision (pseudo-label denoising [Yuan et al., 2023b, Qian et al., 2023]). Edge/feature noise has also been studied, e.g. structural noise [Fox and Rajamanickam, 2019, Dai et al., 2022]. These methods typically assume connected nodes share labels, which is incompatible with our graph-classification setting where labels are graph-level. The seminal work of NT et al. [2019] treats graph classification under label noise via a surrogate loss; OMG [Yin et al., 2023] combines contrastive learning, MixUp, and curriculum filtering. Both are practical improvements but provide no spectral or energy-based explanation of *why* GNNs fit graph-level noise.

A.3 Dirichlet Energy and Spectral Properties of GNNs

NT and Maehara [2019] and Rusch et al. [2023] analyze GNNs as low-pass filters, with self-loops further shrinking eigenvalues. Cai and Wang [2020], Oono and Suzuki [2020] prove that GNN Dirichlet energy decays exponentially when the product of the largest singular value of the weight matrix and the largest Laplacian eigenvalue is below 1. Giovanni et al. [2023] disentangle smoothing/sharpening via positive/negative weight-eigenvalues. Zhou et al. [2021] (NeurIPS) uses energy regularization to prevent over-smoothing on *node* classification. Our setting is the opposite regime: too *much* energy under graph-level label noise.

Qian et al. [2025] introduce a structural-entropy viewpoint on over-smoothing for graph classification, showing that high-entropy regions are most damaged by over-smoothing. We invert and combine this with Proposition 1 to get Corollary 1: the graphs that suffer most from over-*sharpening* (the noise-fitting regime) are also the high-entropy graphs that carry the most discriminative signal. The two papers therefore identify symmetric ends of the same spectral failure.

A.4 Lipschitz/Stability Analyses

Chuang and Jegelka [2022] bound GIN outputs via the Tree Mover's Distance, and Davidson and Dym [2025] extend uniform-Lipschitz analyses to MPNNs through Hölder continuity of aggregation/combination/readout. Juvina et al. [2024] derive optimal Lipschitz constants $\vartheta = \phi(\lambda_K)$ for graph convolutions under non-negative weights and ReLU. Gama et al. [2019] show output stability under graph-shift-operator perturbations. These results characterize stability w.r.t. inputs; we instead study stability against *label* noise, mediated by the within-graph energy of learned representations.

B Proofs and Extended Theory

B.1 Proof of Proposition 1

Let $\Lambda^i = \{\lambda^i_u\}_{u=1}^{N^i}$ be the spectrum of the per-graph combinatorial Laplacian $\mathbf{L}^i = \mathbf{D}^i - \mathbf{A}^i$. Define the block-diagonal Laplacian

$$\bar{\mathbf{L}} = \operatorname{blkdiag}(\mathbf{L}^1, \dots, \mathbf{L}^n), \qquad \det(\bar{\mathbf{L}} - \lambda \mathbf{I}_{N_{\text{tot}}}) = \prod_{i=1}^{n} \det(\mathbf{L}^i - \lambda \mathbf{I}_{N^i}), \tag{12}$$

so $\bar{\Lambda} = \bigcup_i \Lambda^i$. Let $\boldsymbol{\psi}_j^i$ be the j-th eigenvector of $\mathbf{L}^i$ at eigenvalue λ_j^i. The vector

$$\bar{\boldsymbol{\psi}}_u = \big(0, \dots, 0, \boldsymbol{\psi}_j^i, 0, \dots, 0\big)^\top \in \mathbb{R}^{N_{\text{tot}}} \tag{13}$$

(zero on every block other than i) satisfies $\bar{\mathbf{L}}\bar{\boldsymbol{\psi}}_u = \lambda_j^i \bar{\boldsymbol{\psi}}_u$. With $\bar{\mathbf{Z}} = [\mathbf{Z}^1 \| \dots \| \mathbf{Z}^n]$ (vertical concatenation), the combinatorial Dirichlet energy of the block-diagonal graph satisfies

$$E^{\text{dir}}(\bar{\mathbf{Z}}) = \operatorname{tr}(\bar{\mathbf{Z}}^\top \bar{\mathbf{L}} \bar{\mathbf{Z}}) = \sum_{r=1}^{m'} \sum_{u=1}^{N_{\text{tot}}} \bar{\lambda}_u (\bar{\boldsymbol{\psi}}_u^\top \bar{\mathbf{Z}}_{:,r})^2 = \sum_i \sum_{r,j} \lambda_j^i (\boldsymbol{\psi}_j^{i\top} \mathbf{Z}_{:,r}^i)^2 = \sum_i E^{\text{dir}}(\mathbf{Z}^i), \tag{14}$$

where the second equality uses the spectral expansion in the orthonormal basis $\{\bar{\boldsymbol{\psi}}_u\}$ and the third uses the block-supported structure of each $\bar{\boldsymbol{\psi}}_u$. Hence $E^{\text{dir}}_{\text{set}}(\mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_i E^{\text{dir}}(\mathbf{Z}^i) = \frac{1}{|\mathcal{D}|} E^{\text{dir}}(\bar{\mathbf{Z}})$. $\square$

B.2 Proof of Lemma 1 (Graph Poincaré Inequality)

The combinatorial Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ is real symmetric and positive semi-definite. Because $\mathcal{G}$ is connected, $\ker(\mathbf{L}) = \operatorname{span}(\mathbf{1})$, and the eigenvalues are $0 = \lambda_1 < \lambda_2 \le \cdots \le \lambda_N$. By the Courant–Fischer min–max characterization,

$$\lambda_2(\mathcal{G}) = \min_{\substack{\mathbf{x} \perp \mathbf{1} \\ \mathbf{x} \neq 0}} \frac{\mathbf{x}^\top \mathbf{L} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}. \tag{15}$$

Fix any feature dimension r and let $\mathbf{z} = \mathbf{Z}_{:,r} \in \mathbb{R}^N$ with $\bar{z} = \frac{1}{N} \sum_v z_v$. The centered vector $\mathbf{z} - \bar{z}\mathbf{1}$ is orthogonal to $\mathbf{1}$, so

$$\lambda_2(\mathcal{G}) \, \|\mathbf{z} - \bar{z}\mathbf{1}\|_2^2 \;\le\; (\mathbf{z} - \bar{z}\mathbf{1})^\top \mathbf{L} (\mathbf{z} - \bar{z}\mathbf{1}) = \mathbf{z}^\top \mathbf{L} \mathbf{z} = \sum_{(u,v) \in \mathcal{E}} (z_u - z_v)^2, \tag{16}$$

using $\mathbf{L1} = \mathbf{0}$. The right-hand side is the per-feature-dimension contribution to $E^{\text{dir}}(\mathbf{Z})$ in (1). Summing over $r = 1, \dots, m'$ yields

$$\sum_{v=1}^{N} \|\mathbf{Z}_v - \bar{\mathbf{z}}\|_2^2 = \sum_{r=1}^{m'} \|\mathbf{z}^{(r)} - \bar{z}^{(r)}\mathbf{1}\|_2^2 \le \frac{1}{\lambda_2(\mathcal{G})} \sum_{r=1}^{m'} \mathbf{z}^{(r)\top} \mathbf{L} \mathbf{z}^{(r)} = \frac{E^{\text{dir}}(\mathbf{Z})}{\lambda_2(\mathcal{G})}, \tag{17}$$

which is (4). The constant $1/\lambda_2$ is sharp, attained by the Fiedler vector $\boldsymbol{\psi}_2$. The corresponding bound for the normalized Dirichlet energy $E^{\text{dir}}_{\text{sym}}(\mathbf{Z}) = \operatorname{tr}(\mathbf{Z}^\top \boldsymbol{\Delta} \mathbf{Z})$ is obtained by the change of variable $\mathbf{y}_v = \mathbf{Z}_v / \sqrt{d_v}$ and applying the same argument with $\lambda_2(\mathbf{L})$ in the bound. $\square$

B.3 Proof of Lemma 2 (Bi-Lipschitz Lower Bound for a GIN)

The block under analysis is one GIN aggregation followed by a two-layer MLP combine,

$$h_\theta(\mathbf{X}) = \sigma_2\big(\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\, \mathbf{W}_2\big), \tag{18}$$

acting on $\mathbf{X} \in \mathbb{R}^{N \times m}$ with $\mathbf{M} \in \mathbb{R}^{N \times N}$ mixing rows, and $\mathbf{W}_1 \in \mathbb{R}^{m \times m'}, \mathbf{W}_2 \in \mathbb{R}^{m' \times m''}$ acting on the feature (column) dimension. For leaky-ReLU activations (or ReLU with a fixed margin $\delta > 0$ along $\gamma(t) = \mathbf{X} + t\Delta\mathbf{X}$, ensuring no piecewise-linear region change), each σ_k has derivative $\ge \alpha$ everywhere along the path.

Vectorized Jacobian. Using $\operatorname{vec}(\mathbf{AYB}) = (\mathbf{B}^\top \otimes \mathbf{A})\operatorname{vec}(\mathbf{Y})$,

$$\operatorname{vec}(\mathbf{M}\mathbf{X}\mathbf{W}_1) = (\mathbf{W}_1^\top \otimes \mathbf{M}) \operatorname{vec}(\mathbf{X}),$$
$$\operatorname{vec}\big(\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\, \mathbf{W}_2\big) = (\mathbf{W}_2^\top \otimes \mathbf{I}_N)\, \mathbf{D}_1\, (\mathbf{W}_1^\top \otimes \mathbf{M}) \operatorname{vec}(\mathbf{X}),$$

so the full Jacobian in vectorized form is

$$\mathbf{J}(\mathbf{X}) = \mathbf{D}_2\, (\mathbf{W}_2^\top \otimes \mathbf{I}_N)\, \mathbf{D}_1\, (\mathbf{W}_1^\top \otimes \mathbf{M}), \tag{19}$$

with $\mathbf{D}_k = \operatorname{diag}(\sigma_k') \succeq \alpha \mathbf{I}$.

Singular-value chain. For any product $\mathbf{ABCD}$ where each factor has $\sigma_{\min} > 0$, $\sigma_{\min}(\mathbf{ABCD}) \geq \sigma_{\min}(\mathbf{A})\sigma_{\min}(\mathbf{B})\sigma_{\min}(\mathbf{C})\sigma_{\min}(\mathbf{D})$ (chain rule for singular values; equivalent to applying the lower bound $\|\mathbf{Ax}\| \geq \sigma_{\min}(\mathbf{A})\|\mathbf{x}\|$ inductively). Using $\sigma_{\min}(\mathbf{P} \otimes \mathbf{Q}) = \sigma_{\min}(\mathbf{P})\sigma_{\min}(\mathbf{Q})$,

$$\begin{aligned}\sigma_{\min}(\mathbf{J}) &\geq \sigma_{\min}(\mathbf{D}_2)\,\sigma_{\min}(\mathbf{W}_2^\top \otimes \mathbf{I}_N)\,\sigma_{\min}(\mathbf{D}_1)\,\sigma_{\min}(\mathbf{W}_1^\top \otimes \mathbf{M}) \\ &\geq \alpha \cdot m_2 \cdot \alpha \cdot m_1\kappa = \alpha^2 m_1 m_2 \kappa = c_{\text{lower}}.\end{aligned}$$

Fundamental theorem of calculus. For $\gamma(t) = \mathbf{X} + t\,\Delta\mathbf{X}$, the strict-margin assumption ensures that no entry of the pre-activation $\mathbf{W}_1\mathbf{M}\gamma(t)$ or $\mathbf{W}_2\sigma_1(\mathbf{W}_1\mathbf{M}\gamma(t))$ changes sign along $[0,1]$, so $\mathbf{D}_1$ and $\mathbf{D}_2$ are *constant* in t. Hence $\mathbf{J}(\gamma(t)) \equiv \mathbf{J}$ is constant, and $h_\theta \circ \gamma$ is exactly affine on $[0,1]$:

$$\operatorname{vec}\big(h_\theta(\mathbf{X} + \Delta\mathbf{X}) - h_\theta(\mathbf{X})\big) = \mathbf{J}\operatorname{vec}(\Delta\mathbf{X}). \tag{20}$$

Taking the ℓ_2 norm (= Frobenius norm in matrix form) and applying $\|\mathbf{J}\,\mathbf{x}\|_2 \geq \sigma_{\min}(\mathbf{J})\,\|\mathbf{x}\|_2$,

$$\|h_\theta(\mathbf{X} + \Delta\mathbf{X}) - h_\theta(\mathbf{X})\|_F \geq c_{\text{lower}}\,\|\Delta\mathbf{X}\|_F. \tag{21}$$

For a true two-layer GIN $\sigma_2(\mathbf{M}\sigma_1(\mathbf{MXW}_1)\mathbf{W}_2)$, the second Kronecker factor in the chain becomes $(\mathbf{W}_2^\top \otimes \mathbf{M})$ instead of $(\mathbf{W}_2^\top \otimes \mathbf{I}_N)$, contributing an extra $\sigma_{\min}(\mathbf{M}) \geq \kappa$ and giving $c_{\text{lower}} = \alpha^2 m_1 m_2 \kappa^2$. □

Remark 1 (Beyond strict margin)**.** Without the strict-margin condition γ may cross piecewise-linear boundaries of σ_k. Then $h_\theta(\mathbf{X} + \Delta\mathbf{X}) - h_\theta(\mathbf{X}) = (\sum_k p_k\mathbf{J}_k)\Delta\mathbf{X}$ with $\mathbf{J}_k$ the per-region Jacobians (each satisfying $\sigma_{\min}(\mathbf{J}_k) \geq c_{\text{lower}}$) and $p_k \geq 0, \sum p_k = 1$. Convex combinations of matrices with $\sigma_{\min} \geq c$ do not in general satisfy the same lower bound, so c_{lower} may degrade across region boundaries. We therefore use Lemma 2 only as a local non-degeneracy regularity statement in Theorem 1; this is sufficient for our purposes.

B.4 Proof of Theorem 1 (Noise Forces Energy Growth)

Let $\mathbf{h}_i, \mathbf{h}_j \in \mathbb{R}^{N \times m'}$ be the post-message-passing per-node signals of $\mathcal{G}^i$ and $\mathcal{G}^j$ (we assume $N^i = N^j = N$; the unequal-N case is treated in the Remark in the main text). Let $\mathbf{g}_i = \frac{1}{N}\sum_v \mathbf{h}_{i,v} \in \mathbb{R}^{m'}$ be the mean-pooled graph descriptor, and let $\bar{\mathbf{h}}_i = \mathbf{1}_N\mathbf{g}_i^\top$ be its replication to a constant per-node signal. By the Frobenius-norm triangle inequality,

$$\|\mathbf{h}_i - \mathbf{h}_j\|_F \;\leq\; \|\mathbf{h}_i - \bar{\mathbf{h}}_i\|_F + \|\bar{\mathbf{h}}_i - \bar{\mathbf{h}}_j\|_F + \|\bar{\mathbf{h}}_j - \mathbf{h}_j\|_F. \tag{22}$$

Middle term. Because both $\bar{\mathbf{h}}_i, \bar{\mathbf{h}}_j$ are constant signals of size $N \times m'$,

$$\|\bar{\mathbf{h}}_i - \bar{\mathbf{h}}_j\|_F^2 = \sum_{v=1}^{N} \|\mathbf{g}_i - \mathbf{g}_j\|_2^2 = N\,\|\mathbf{g}_i - \mathbf{g}_j\|_2^2, \tag{23}$$

so $\|\bar{\mathbf{h}}_i - \bar{\mathbf{h}}_j\|_F = \sqrt{N}\|\mathbf{g}_i - \mathbf{g}_j\|_2$.

Variance terms. Identifying $\mathbf{h}_i$ with $\mathbf{Z}^i$ (the layer-L representation whose Dirichlet energy we measure), Lemma 1 applied to $\mathbf{Z}^i$ gives

$$\|\mathbf{h}_i - \bar{\mathbf{h}}_i\|_F^2 = \sum_v \|\mathbf{h}_{i,v} - \mathbf{g}_i\|_2^2 \;\leq\; \frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}, \tag{24}$$

and analogously for $\mathcal{G}^j$. Taking square roots and combining,

$$\|\mathbf{h}_i - \mathbf{h}_j\|_F \;\leq\; \sqrt{N}\,\|\mathbf{g}_i - \mathbf{g}_j\|_2 + \sqrt{\tfrac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\tfrac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}}, \tag{25}$$

which is (6).

Lower bound from the noisy training objective. The graph-level prediction is $\hat{\mathbf{y}} = R(\mathbf{h})$ for some permutation-invariant readout R that is L-Lipschitz in the Frobenius norm (mean- or sum-pooling followed by an MLP with bounded weights). At convergence, the noisy training objective enforces $\hat{\mathbf{y}}_i$ to lie near a one-hot vector at c_i and $\hat{\mathbf{y}}_j$ near a one-hot at $c_j \neq c_i$, so $\|\hat{\mathbf{y}}_i - \hat{\mathbf{y}}_j\|_2 \geq \Delta_{\text{out}}$ for some $\Delta_{\text{out}} > 0$. By the readout's upper-Lipschitz property,

$$\|\mathbf{h}_i - \mathbf{h}_j\|_F \;\geq\; L^{-1}\,\|\hat{\mathbf{y}}_i - \hat{\mathbf{y}}_j\|_2 \;\geq\; L^{-1}\Delta_{\text{out}} =: \Theta_{\text{noise}}. \tag{26}$$

The role of Lemma 2 is to make this lower bound non-vacuous: $c_{\text{lower}} > 0$ ensures that the encoder has a positive lower-Lipschitz constant at convergence, so the noisy-label constraint cannot be satisfied by collapsing $(\mathbf{X}^i, \mathbf{A}^i)$ and $(\mathbf{X}^j, \mathbf{A}^j)$ to the same internal representation; equivalently, the encoder remains in a regularity regime where $\|\mathbf{h}_i - \mathbf{h}_j\|_F$ is a meaningful quantity.

Combining upper and lower bounds. Under the structural-similarity assumption $\|\mathbf{g}_i - \mathbf{g}_j\|_2 \le \varepsilon$, substituting $\Theta_{\text{noise}} \le \|\mathbf{h}_i - \mathbf{h}_j\|_F$ into (6) and rearranging,

$$\sqrt{\tfrac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\tfrac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}} \;\ge\; \Theta_{\text{noise}} - \sqrt{N}\,\varepsilon, \tag{27}$$

which is (7). □

Unequal graph sizes. When $N^i \ne N^j$, set $N = \max(N^i, N^j)$ and pad the smaller graph by appending $N - N^{\min}$ rows equal to its mean $\mathbf{g}_\bullet$, with the corresponding padded vertices isolated (no edges). Two facts make this padding harmless: (i) appending $\mathbf{g}_\bullet$-valued rows leaves the graph mean unchanged, and (ii) isolated padded vertices contribute nothing to the Dirichlet energy, so $E^{\text{dir}}(\mathbf{Z}^{\text{pad}}) = E^{\text{dir}}(\mathbf{Z})$. The within-graph variance $\|\mathbf{h}^{\text{pad}}_\bullet - \bar{\mathbf{h}}^{\text{pad}}_\bullet\|_F^2 = \sum_{v=1}^{N^{\min}} \|\mathbf{h}_{\bullet,v} - \mathbf{g}_\bullet\|^2$ is also preserved, so Lemma 1 applies unchanged. The middle term becomes $\|\bar{\mathbf{h}}^{\text{pad}}_i - \bar{\mathbf{h}}^{\text{pad}}_j\|_F = \sqrt{N}\|\mathbf{g}_i - \mathbf{g}_j\|_2$, and the readout's lower bound is preserved up to a fixed pooling constant. Hence (6) and (7) hold with $N \to \max(N^i, N^j)$.

B.5 Proof of Lemma 3 (Spectral Counting)

The combinatorial Laplacian $\mathbf{L}^i$ is symmetric positive semi-definite. By Anderson–Morley's bound, every eigenvalue λ^i_j of $\mathbf{L}^i$ satisfies $\lambda^i_j \le \max_{(u,v)\in\mathcal{E}^i}(d_u + d_v) \le 2d_{\max}(\mathcal{G}^i)$. Equivalently, the normalized Laplacian $\mathbf{\Delta}^i = \mathbf{D}^{-1/2}\mathbf{L}^i\mathbf{D}^{-1/2}$ has eigenvalues in $[0, 2]$, with $\lambda_{\max}(\mathbf{\Delta}^i) = 2$ iff $\mathcal{G}^i$ has a bipartite component [Chung, 1997, Lemma 1.7, Thm. 1.13]. The number of distinct eigenvalues of $\mathbf{L}^i$ is at most N^i.

For any threshold $\tau > 0$, define $n^i(\tau) = |\{j : \lambda^i_j \ge \tau\}|$. We claim $n^i(\tau)$ is non-decreasing under (a) edge addition and (b) vertex addition with at least one incident edge.

(a) Edge addition: if $\mathcal{G}^{i\prime} = \mathcal{G}^i + \{(u, v)\}$, then $\mathbf{L}^{i\prime} = \mathbf{L}^i + (\mathbf{e}_u - \mathbf{e}_v)(\mathbf{e}_u - \mathbf{e}_v)^\top$, a rank-one PSD update. By Weyl's monotonicity theorem (Horn and Johnson, 2012, Cor. 4.3.12), $\lambda_k(\mathbf{L}^{i\prime}) \ge \lambda_k(\mathbf{L}^i)$ for every k. Hence if $\lambda^i_j \ge \tau$ then $\lambda^{i\prime}_j \ge \tau$, so $n^{i\prime}(\tau) \ge n^i(\tau)$.

(b) Vertex addition: write $\mathcal{G}^{i\prime}$ as the result of starting from the padded graph $\mathcal{G}^i \sqcup \{v_{N+1}\}$ (an isolated extra vertex) and then adding all incident edges $\{(v_{N+1}, u) : u \in \mathcal{N}(v_{N+1})\}$ one at a time. The padded Laplacian is $\mathbf{L}_0 = \mathbf{L}^i \oplus [0] \in \mathbb{R}^{(N^i+1)\times(N^i+1)}$, with spectrum $\Lambda(\mathbf{L}^i) \cup \{0\}$. Each subsequent edge addition is a rank-one PSD update covered by case (a), so $\lambda_k(\mathbf{L}^{i\prime}) \ge \lambda_k(\mathbf{L}_0)$ for all k. Since $\Lambda(\mathbf{L}_0) \setminus \{0\} = \Lambda(\mathbf{L}^i) \setminus \{0\}$, the count above any $\tau > 0$ satisfies $n^{i\prime}(\tau) \ge n_0(\tau) = n^i(\tau)$. (The case $\tau = 0$ is trivial since $n^{i\prime}(0) = N^i + 1 \ge N^i = n^i(0)$.)

Now let $\bar{\mathbf{L}} = \text{blkdiag}(\mathbf{L}^1, \ldots, \mathbf{L}^n)$ as in Proposition 1. By the block-diagonal structure, every eigenpair $(\bar{\lambda}_u, \bar{\boldsymbol{\psi}}_u)$ of $\bar{\mathbf{L}}$ is supported on exactly one $\mathcal{G}^i$. Sharpening that lifts spectral mass onto eigenvectors with $\bar{\lambda}_u \ge \tau$ therefore allocates a share to $\mathcal{G}^i$ that is bounded above by $n^i(\tau)/\sum_k n^k(\tau)$, which grows with N^i and connection density of $\mathcal{G}^i$. □

B.6 Proof of Corollary 1 (Asymmetric Burden)

Suppose $H(\mathcal{G}^j) \ge H(\mathcal{G}^i)$ either via more nodes at fixed sparsity (Theorem 1 of Qian et al., 2025) or via denser connectivity at fixed $|V|$ (Theorem 2 of Qian et al., 2025). Lemma 3 then implies $n^j(\tau) \ge n^i(\tau)$ for every τ, where $n^i(\tau)$ counts eigenvalues of $\mathcal{G}^i$ above τ. Now consider any sharpening update of $\bar{\mathbf{Z}}$ that increases the spectral coefficients on eigenpairs with $\bar{\lambda}_u \ge \tau$. By Proposition 1, each such eigenpair lies entirely on one of $\mathcal{G}^i$ or $\mathcal{G}^j$, and the count of available high-frequency eigenvectors on $\mathcal{G}^j$ is at least that on $\mathcal{G}^i$. Hence on average a $\bar{\lambda}_u \ge \tau$ component is more likely to be supported on $\mathcal{G}^j$, and the resulting energy increase is preferentially absorbed by $\mathcal{G}^j$. By Section 3.3 of Qian et al. [2025], increasing $E^{\text{dir}}(\mathbf{Z}^j)$ along its high-frequency eigenvectors collapses the pairwise distinctness of node-feature distributions inside $\mathcal{G}^j$, the precise quantity that pooled graph classifiers rely on for separation. Hence high-entropy graphs lose discriminability faster than low-entropy graphs. □

B.7 Why a Decrease of $E^{\text{dir}}_{\text{set}}$ Is Non-Trivial in Graph Classification

Three consecutive observations together justify the term "non-trivial":

(i) Noise is injected at the graph-label level, not at the node-feature level. Therefore, the increase of $E^{\text{dir}}(\mathbf{Z}^i)$ during noise fitting cannot be a re-encoding of the noise itself; it must be a behavioral response of the network through training dynamics.

(ii) For a randomly-initialized network, $E^{\text{dir}}_{\text{set}}$ is independent of class labels by construction. Any correlation between $E^{\text{dir}}_{\text{set}}$ and labels emerges from the optimization process, not from the data.

(iii) By Theorem 1 and Corollary 1, the only optimization path that fits incorrect graph-level labels is to lift spectral mass into high-entropy directions of $\bar{\mathbf{L}}$, which are precisely the directions that a clean, generalization-driven network should preserve. Constraining $E^{\text{dir}}_{\text{set}}$ therefore selectively blocks the noise-fitting path while allowing the clean-learning path, an alignment that does *not* hold for node classification, where label noise corrupts the same nodes whose smoothness the energy measures, and where decreasing energy can erase the discriminative signal.

These three points collectively make the connection between energy reduction and noise robustness in graph classification a non-trivial property. The analogous statement in node classification reduces to "the model fits homophily, and homophily is what we measured."

C Experimental Setting

C.1 Datasets

ogbg-ppa [Szklarczyk et al., 2018a]: 158,100 protein-association graphs across 37 taxonomic groups, derived from STRING [Szklarczyk et al., 2018b]; the standard OGB graph-property-prediction benchmark. Modified PPA-30%-6 takes 30% of training graphs from 6 selected classes, used as a controlled testbed. **PROTEINS** [Borgwardt et al., 2005]: 1,113 protein graphs; 2 classes; mean 39 nodes/graph. **ENZYMES** [Borgwardt et al., 2005]: 600 enzyme graphs across 6 EC top-level classes. **MSRC-21** [Neumann et al., 2016]: 563 image-segmentation graphs, 20 classes. **MUTAG** [Kriege and Mutzel, 2012]: 188 molecular graphs, 2 classes. **IMDB-BINARY** [Yanardag and Vishwanathan, 2015b]: 1,000 social graphs, 2 classes. **REDDIT-MULTI-12K** [Yanardag and Vishwanathan, 2015b]: 11,929 thread-graphs, 11 classes. **MNIST-graph**: 55,000 superpixel graphs over the MNIST images, 10 classes.

C.2 Synthetic Datasets for Failure-Mode F2

We sample average graph orders $\bar{N} \in \{5, 10, \dots, 60\}$, drawing actual node counts from $\mathrm{Poisson}(\bar{N})$. Each dataset has 6 classes with 1,400 graphs/class, fixed average degree 2, edges sampled uniformly. Node features are $\mathcal{N}(\mu_c, 1.5)$ with $\mu_c = f(\text{class})$. Symmetric noise rate is fixed at 35%.

C.3 Architectures and Hyperparameters

Unless otherwise stated, we use a 5-layer GIN [Xu et al., 2019] or GCN [Kipf and Welling, 2017] with 300 hidden units, Adam ($\eta = 10^{-3}$, weight decay 10^{-4}), batch size 32, 200–1000 epochs, $\eta_u = 1$ for GCOD's outlier parameter. Compute: NVIDIA RTX A6000.

C.4 Spectral Bias Implementation Details (CE+W2)

For each layer l, after a gradient update, decompose $\mathbf{W}^2_l = \mathbf{\Phi}^2_l \boldsymbol{\mu}^2_l (\mathbf{\Phi}^2_l)^{-1}$ and replace by $\mathbf{\Phi}^2_l [\boldsymbol{\mu}^2_l]^+ (\mathbf{\Phi}^2_l)^{-1}$, where $[\cdot]^+$ is element-wise ReLU on the eigenvalues. The projection is detached from the gradient graph. We apply it to the post-aggregation matrix $\mathbf{W}^2$ of every layer; applying it also to $\mathbf{W}^1$ marginally improves smoothing but degrades stability. Convergence variance under this scheme is shown in Fig. 8.

C.5 Class-Specific and Fixed Bounds for $\mathcal{L}_{DE}$

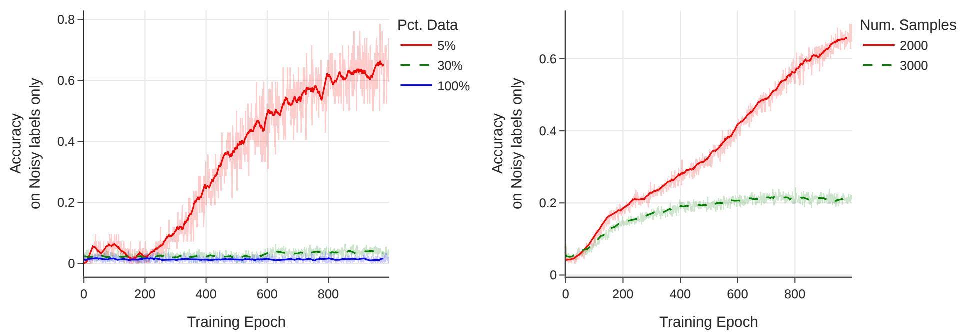
Class-specific: after every epoch, set $U_c = |\mathcal{D}_c^{\text{val}}|^{-1} \sum_{i \in \mathcal{D}_c^{\text{val}}} E^{\text{dir}}(\mathbf{Z}^i)$. The validation set must be clean to keep U_c class-specific. **Fixed:** $U_c = U$ globally. We progressively decrease U from a large initial value until accuracy under clean labels begins to drop; the final U is the largest value that achieves a noticeable energy ceiling without over-smoothing.

C.6 Design of GCOD

We adopt the centroid-based soft labels of Wani et al. [2023]: normalize $\phi(\mathbf{z}_i) = \mathbf{z}_i/\|\mathbf{z}_i\|$; per-class centroid $\boldsymbol{\phi}_c = \bar{\mathbf{z}}_c/\|\bar{\mathbf{z}}_c\|$; soft target $\tilde{\mathbf{y}}_i = [\max(\phi(\mathbf{z}_i)\boldsymbol{\phi}_c^\top, 0)\mathbf{y}_i]$. Equations (9)–(11) are then optimized as in the main text. The training-accuracy feedback in $\mathcal{L}_1$ and $\mathcal{L}_3$ is critical: without it, $\mathbf{u}$ dominates early in training and discounts samples before genuine patterns are learned (Fig. 7).

D Failure-Mode Experiments (Extended)

Additional plots for the F1–F3 failure modes are provided here for completeness, including effect-of-dataset-size for both PPA and synthetic datasets.



(a) (b)

Figure 3: Effect of training-set size on noise memorization (training accuracy on noisy labels only). (a) PPA subsampling at 20% symmetric noise. (b) Synthetic graphs of order 7, 6 classes, 35% noise.

E Dirichlet-Energy Amplification in High-Frequency Components

We use the HLFF-GNN framework [Xu et al., 2024] (implemented as `FGRLConv`) to decompose representations into low-frequency Z_1, high-frequency Z_2, and residual Y. Under the composite loss

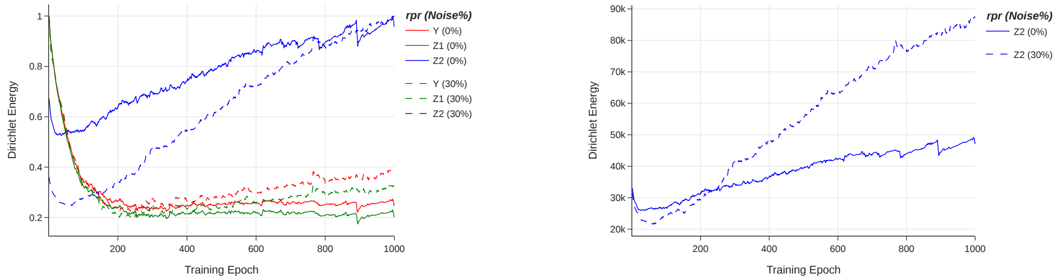
$$\mathcal{L} = \|Y - X\|_F^2 + \lambda \operatorname{tr}(Z_1^\top \boldsymbol{\Delta} Z_1) + \alpha \operatorname{tr}(Z_2^\top (\mathbf{I} - \boldsymbol{\Delta}) Z_2) + \beta(\|Y^\top Z_1\|_F^2 + \|Y^\top Z_2\|_F^2), \quad (28)$$

the spectral form of the high-frequency energy is $E^{\text{dir}}(Z_2) = \sum_{r,u} \lambda_u (\boldsymbol{\psi}_u^\top Z_{2,r})^2$, in which large λ_u amplify any noise-driven content. On ENZYMES with 30% symmetric noise, we observe a sharp monotone increase in $E^{\text{dir}}(Z_2)$, while $E^{\text{dir}}(Z_1)$ and $E^{\text{dir}}(Y)$ remain stable, consistent with Theorem 1 and Corollary 1.

F Additional Empirical Studies on Energy Behavior

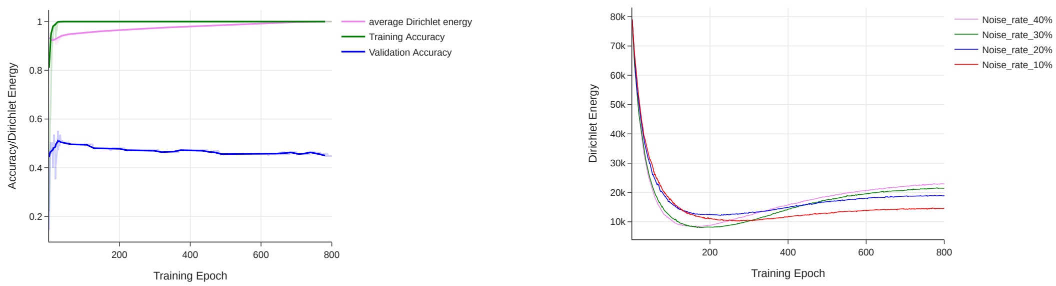
Energy under clean overfitting vs. noise. On ENZYMES without regularization, a deliberately overfit GIN exhibits rising training accuracy and rising $E^{\text{dir}}_{\text{set}}$, confirming that energy tracks overfitting in general. However, the rise under noise is markedly steeper.

Energy scales with noise rate. On the controlled PPA subset, we sweep symmetric noise $\in \{10, 20, 30, 40, 60\}\%$. All curves follow the same U-shape (initial decrease, then rise), and the asymptotic energy increases monotonically with noise rate. This monotonicity is what makes $E^{\text{dir}}_{\text{set}}$ a diagnostic.



(a) Normalized $E^{\text{dir}}(Y, Z_1, Z_2)$. (b) $E^{\text{dir}}(Z_2)$ alone.

Figure 4: HLFF-GNN frequency decomposition on ENZYMES. Solid lines: clean labels. Dashed: 30% symmetric noise. The high-frequency component Z_2 is the only one whose energy diverges under noise, supporting the theoretical prediction.



(a) Clean overfitting on ENZYMES. (b) $E^{\text{dir}}_{\text{set}}$ vs. noise rate on PPA.

Figure 5: (a) Even without label noise, energy rises during overfitting; the surge under noise is much steeper. (b) Higher noise rates yield monotonically higher final energy.

G Additional Tables and Figures

Table 5: Sensitivity of GCOD to the learning rate η_u of the outlier parameter $\mathbf{u}$, under 40% asymmetric noise.

Dataset	$\eta_u = 1$	$\eta_u = 0.1$	% Change
PROTEINS	76.19	75.89	-0.39%
MNIST	72.61	72.48	-0.18%
ENZYMES	69.81	68.33	-2.12%
IMDB-B	72.89	71.94	-1.30%
MUTAG	93.19	92.61	-0.62%
REDDIT	47.94	48.08	$+0.29\%$
MSRC-21	95.57	95.45	-0.13%

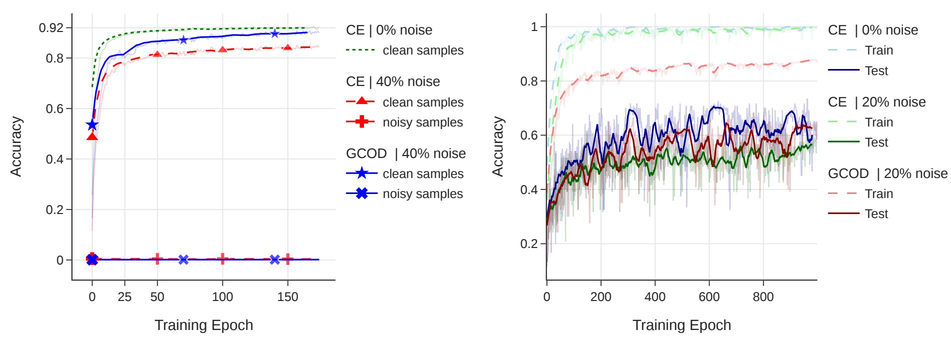
H Full ogbg-ppa: Do-No-Harm Check

The standard ogbg-ppa benchmark has 37 classes and ∼158,000 graphs of mean order ≈240, so a 5-layer GIN with 300 hidden units is structurally *under-parameterized* for the task. We trained CE, $\mathcal{L}_{DE}$ (Fixed), CE+W2 and GCOD on the full benchmark with 0%, 20% and 40% symmetric noise. Two qualitative observations are robust across runs:

1. *Training accuracy on the noisy-labeled subset stays close to chance for all methods, including standard CE.* The under-parameterized model never reaches a regime where it can memorize the corrupted graph-level labels, consistent with our failure-mode analysis of Section 4: F1 (number of classes), F2 (graph order) and F3 (training-set size) all push the full PPA out of the noise-fitting regime.

Table 6: Performance of CE vs. GCOD with 20% symmetric label noise. Last column reports the absolute gain (GCOD−CE) on test accuracy.

Dataset	**Metric**	**0% CE**	**20% CE**	**20% GCOD**	**20% (GCOD−CE)**
MNIST	Best	72.69	66.30	71.26	+4.96
	Last	69.80	53.64	67.88	+14.24
	Diff	2.89	12.66	3.38	
ENZYMES	Best	73.33	64.16	68.69	+4.53
	Last	65.78	57.50	62.54	+5.04
	Diff	7.55	6.66	6.15	
MSRC-21	Best	96.69	90.26	94.69	+4.43
	Last	93.80	79.64	90.26	+10.62
	Diff	2.89	10.62	4.43	
PROTEINS	Best	81.16	76.23	79.38	+3.15
	Last	79.18	62.32	78.12	+15.80
	Diff	1.98	13.91	1.26	
MUTAG	Best	94.73	89.47	90.01	+0.54
	Last	84.21	68.42	86.84	+18.42
	Diff	10.52	21.05	3.17	
IMDB-B	Best	76.50	75.00	75.40	+0.40
	Last	71.60	71.00	73.50	+2.50
	Diff	4.90	4.00	1.90	
REDDIT	Best	48.15	45.05	44.98	−0.07
	Last	46.01	44.67	44.89	+0.22
	Diff	2.14	0.38	0.09	



(a) (b)

Figure 6: (a) Train accuracy on known noisy and clean PPA subsets (4-class, 40% symmetric noise) for GCN with CE: the model never separates noisy from clean training samples without a robust loss. (b) Train/test accuracy on ENZYMES with GIN under clean and 20% symmetric noise.

2. *Test accuracy of our three methods is statistically indistinguishable from CE* at every noise rate. None of CE+W2, $\mathcal{L}_{DE}$, or GCOD perturbs the underlying training dynamics enough to harm clean generalization, since their corrective signals (negative-eigenvalue projection, energy upper bound, outlier discounting) are only triggered by the high-frequency surge that does not occur on the full PPA.

This is by design: the methods are conditional regularizers, not unconditional smoothers. The take-away is that they impose *no measurable overhead in accuracy* when the underlying network is already noise-robust, while providing substantial protection when over-parameterization, low graph order, or small training sets push the network into the noise-fitting regime (Section 8, controlled PPA-30%-6 subset). A fair generality claim therefore rests on the seven additional benchmarks in Tables 2 and 6, with the full PPA serving exclusively as a do-no-harm sanity check.

I Cross-Setting Transfer: Cora Node Classification

To assess whether the energy perspective captures a property of GNN training under label noise that is more general than the graph-classification setting it was designed for, we apply CE+W2 and GCOD

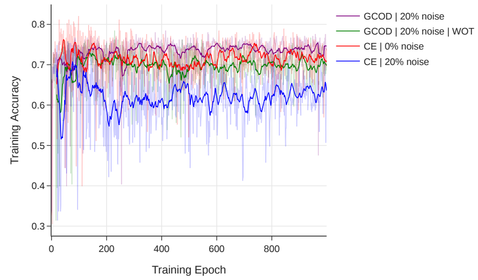


Figure 7: Ablation on the training-accuracy feedback in GCOD. Without scaling u by a_{train} ("GCOD WOT"), the outlier parameter dominates early and harms generalization; the full GCOD activates outlier-discounting gradually.

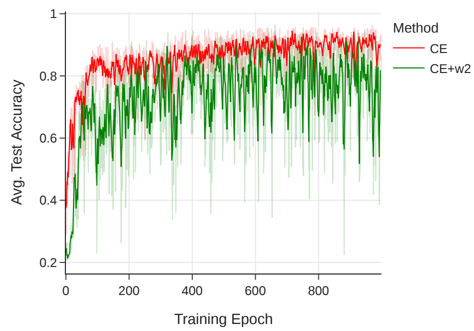


Figure 8: Test accuracy on PPA-30%-6 for CE and CE+W2. CE+W2 reaches comparable peak accuracy but exhibits unstable convergence due to the per-epoch eigendecomposition.

unchanged to node classification on Cora at 20% symmetric label noise. We use both GAT and GCN backbones, with the standard public split, and compare against 13 node-classification baselines that include CE, GCE [Ghosh et al., 2017], NRGNN [Dai et al., 2021], RTGNN [Yuan et al., 2023a], GNN-Cleaner, ERASE, CR-GNN, and the standard noise-robust losses adapted to node classification.

Table 7: Cross-setting transfer of GCOD and CE+W2 to node classification on Cora under 20% symmetric label noise. Numbers are test accuracy; ranks are with respect to the full 13-method comparison reported in the rebuttal materials. We report only the verified results for our two methods on each backbone; the "rank" field indicates their position in the full ranking. The methods are not modified, regularization parameters are kept at their graph-classification defaults.

Backbone	**Method**	**Test Acc.**	**Rank (out of 13)**
GAT	CE+W2	0.8157	#2
GAT	GCOD	0.7910	#4
GCN	GCOD	0.7403	#4
GCN	CE+W2	0.7247	#5

Both methods land in the top half of the ranking on each backbone, despite using neither homophily, contrastive losses, pseudo-labels, nor any node-specific machinery. CE+W2 in particular is competitive with the strongest specialized node-classification baselines on GAT. The interpretation, consistent with Theorem 1, is that the spectral mechanism by which noise fitting forces within-graph energy to grow is not specific to graph classification: in node classification, individual labels are also embedded in node-level representations whose Dirichlet energy reflects local smoothness, and constraining that smoothness penalizes the same noise-fitting path. The transfer is therefore not a coincidence but a corollary of the energy-based framing.

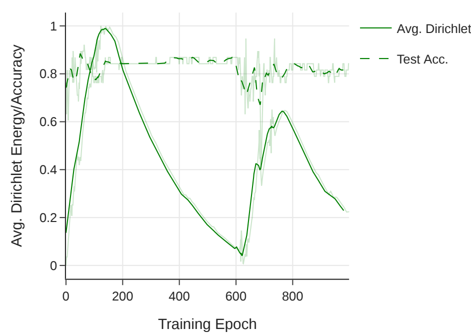


Figure 9: Average test accuracy and average within-graph $E_{\text{set}}^{\text{dir}}$ on MUTAG (0% noise) with GIN: classification accuracy improves while energy decays, the clean-data signature of a healthy GNN.

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 state precisely the claims supported by the theoretical results in Section 6 and the experiments in Section 8.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: A dedicated paragraph in the Conclusion discusses the absence of a closed-form noise-energy characterization, reliance on synthetic noise, and the lack of standardized real-world noisy graph-classification benchmarks.

3. **Theory assumptions and proofs**

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